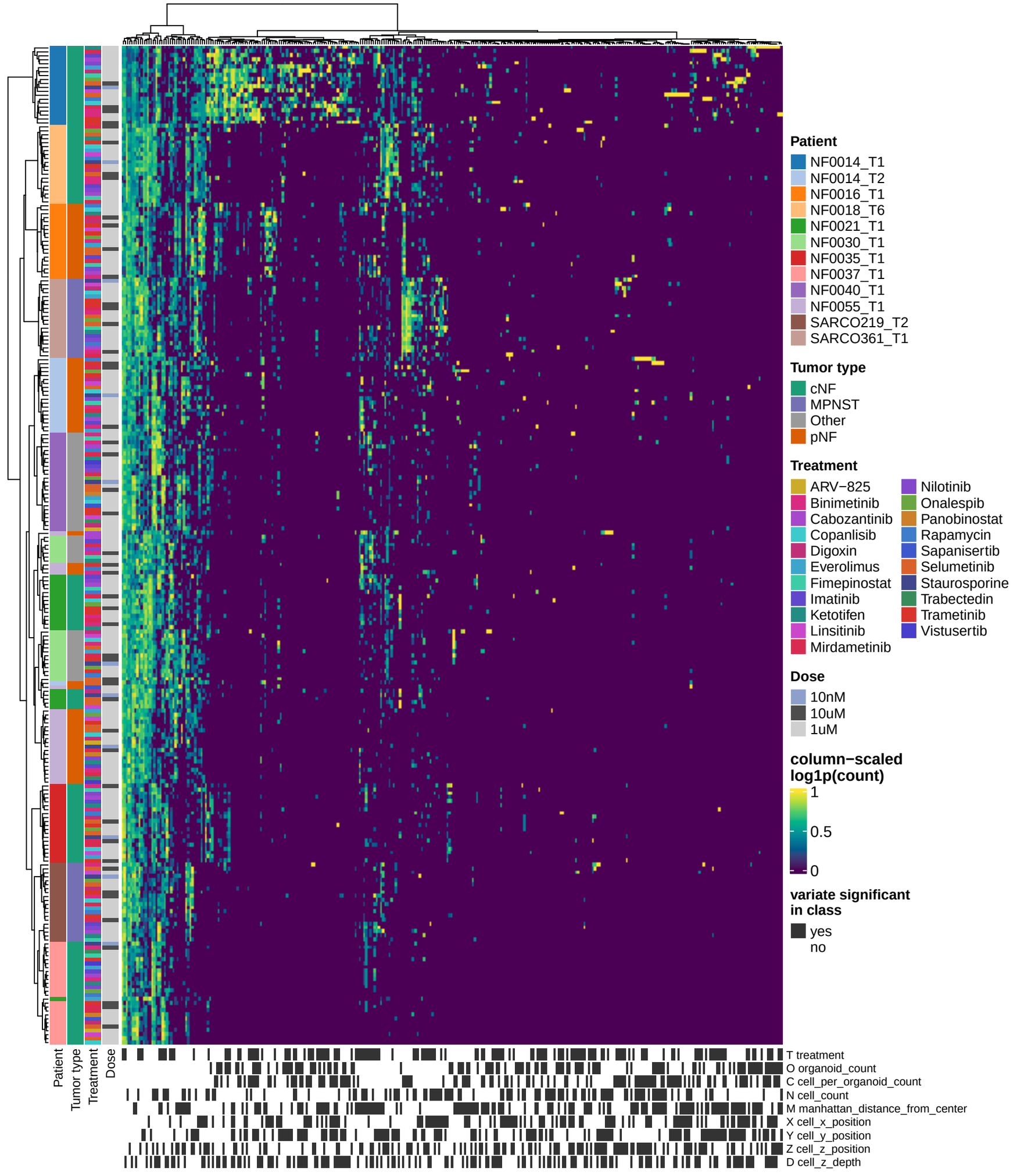
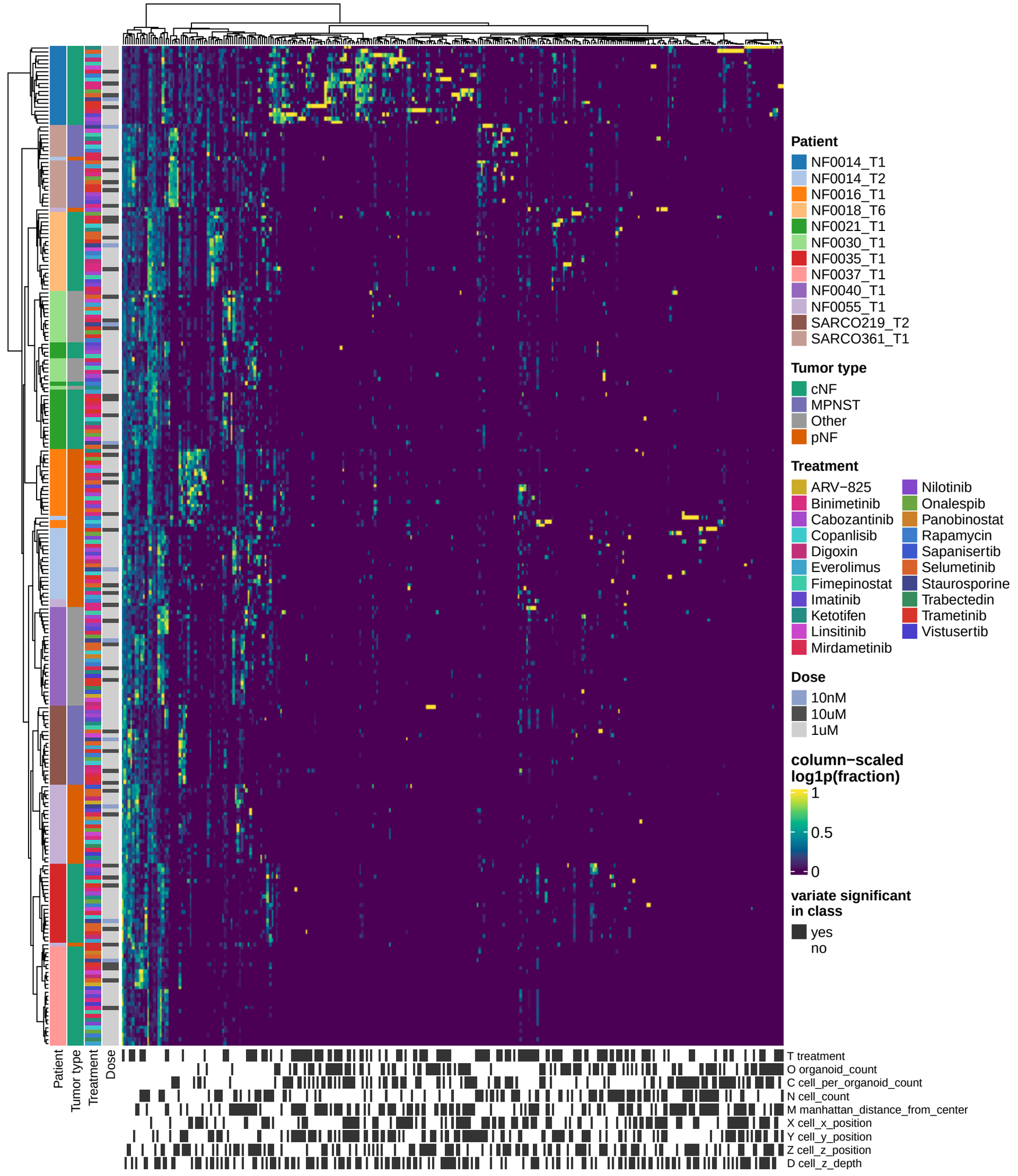
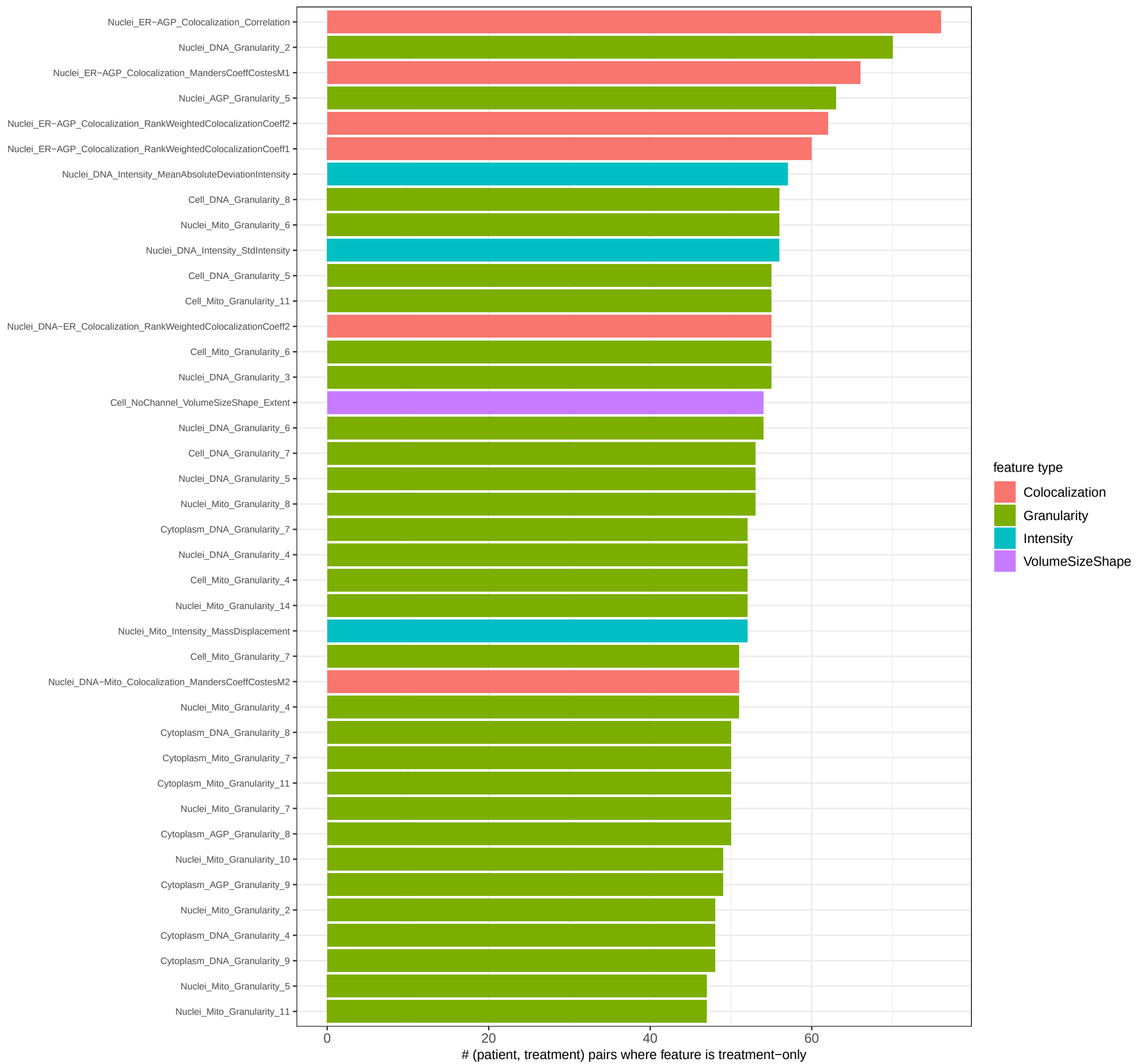
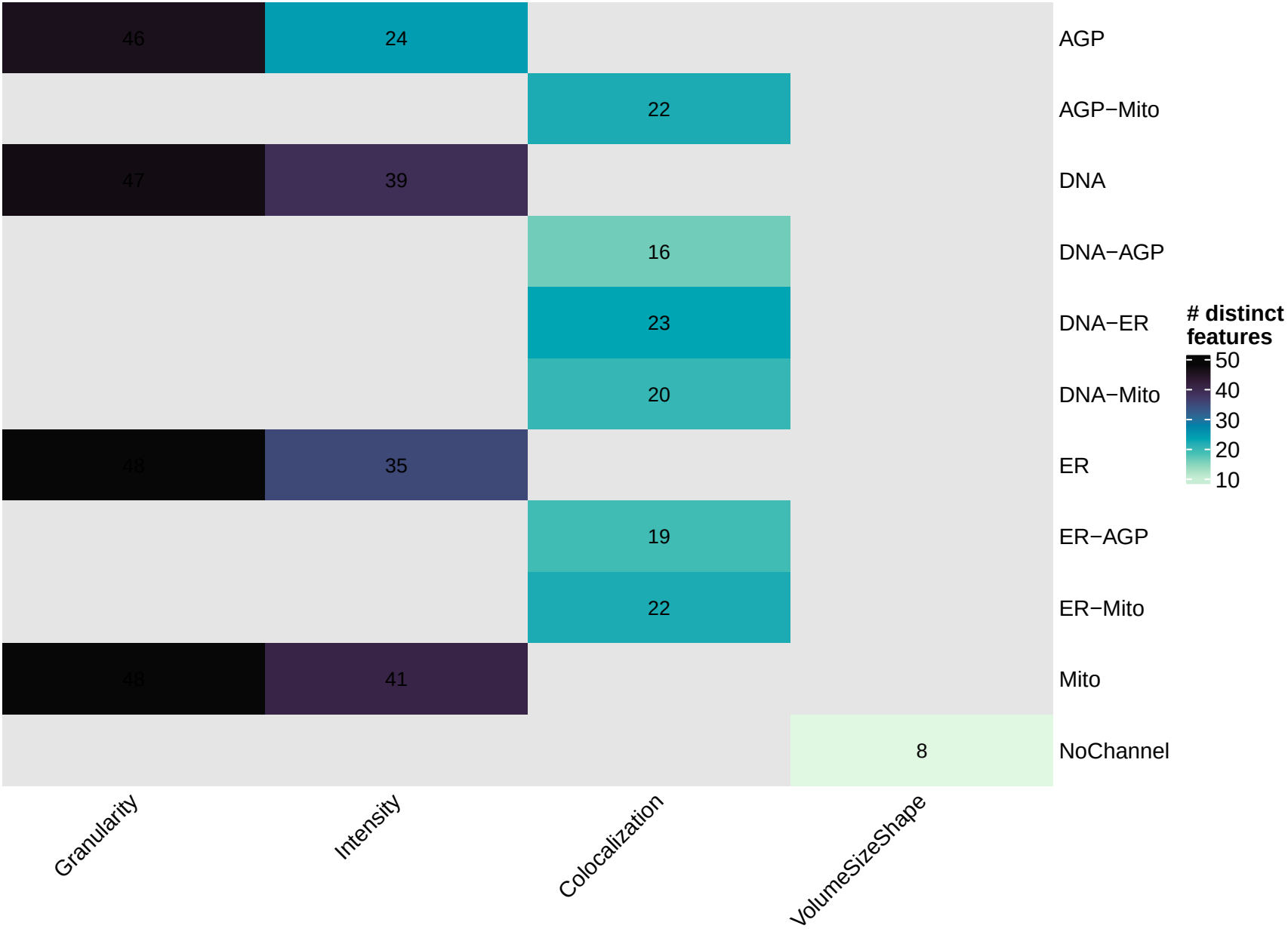

Patient
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
Tumor type
cNF
MPNST
Other
pNF
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
10nM
10uM
1uM
column−scaled log1p(count)
1
0.5
0
variate significant in class
yes
no
Patient
Tumor type
Treatment
Dose
T treatment
O organoid_count
C cell_per_organoid_count
N cell_count
M manhattan_distance_from_center
X cell_x_position
Y cell_y_position
Z cell_z_position
D cell_z_depth


Patient
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
Tumor type
cNF
MPNST
Other
pNF
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
Dose
10nM
10uM
1uM
column−scaled log1p(fraction)
1
0.5
0
variate significant in class
yes
no
Patient
Tumor type
Treatment
Dose
T treatment
O organoid_count
C cell_per_organoid_count
N cell_count
M manhattan_distance_from_center
X cell_x_position
Y cell_y_position
Z cell_z_position
D cell_z_depth

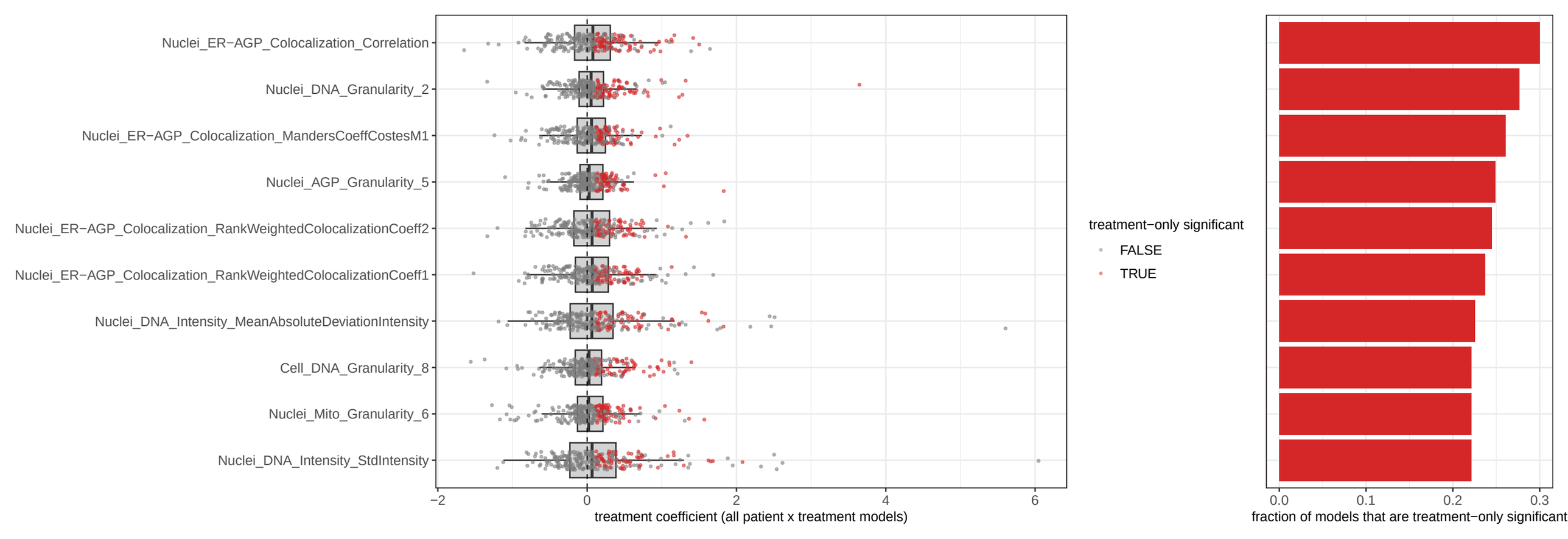

Nuclei_ER−AGP_Colocalization_Correlation
Nuclei_DNA_Granularity_2
Nuclei_ER−AGP_Colocalization_MandersCoeffCostesM1
Nuclei_AGP_Granularity_5
Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff2
Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff1
Nuclei_DNA_Intensity_MeanAbsoluteDeviationIntensity
Cell_DNA_Granularity_8
Nuclei_Mito_Granularity_6
Nuclei_DNA_Intensity_StdIntensity
Cell_DNA_Granularity_5
Cell_Mito_Granularity_11
Nuclei_DNA−ER_Colocalization_RankWeightedColocalizationCoeff2
Cell_Mito_Granularity_6
Nuclei_DNA_Granularity_3
Cell_NoChannel_VolumeSizeShape_Extent
Nuclei_DNA_Granularity_6
Cell_DNA_Granularity_7
Nuclei_DNA_Granularity_5
Nuclei_Mito_Granularity_8
Cytoplasm_DNA_Granularity_7
Nuclei_DNA_Granularity_4
Cell_Mito_Granularity_4
Nuclei_Mito_Granularity_14
Nuclei_Mito_Intensity_MassDisplacement
Cell_Mito_Granularity_7
Nuclei_DNA−Mito_Colocalization_MandersCoeffCostesM2
Nuclei_Mito_Granularity_4
Cytoplasm_DNA_Granularity_8
Cytoplasm_Mito_Granularity_7
Cytoplasm_Mito_Granularity_11
Nuclei_Mito_Granularity_7
Cytoplasm_AGP_Granularity_8
Nuclei_Mito_Granularity_10
Cytoplasm_AGP_Granularity_9
Nuclei_Mito_Granularity_2
Cytoplasm_DNA_Granularity_4
Cytoplasm_DNA_Granularity_9
Nuclei_Mito_Granularity_5
Nuclei_Mito_Granularity_11
0
20
40
60
(patient, treatment) pairs where feature is treatment−only
feature type
Colocalization
Granularity
Intensity
VolumeSizeShape

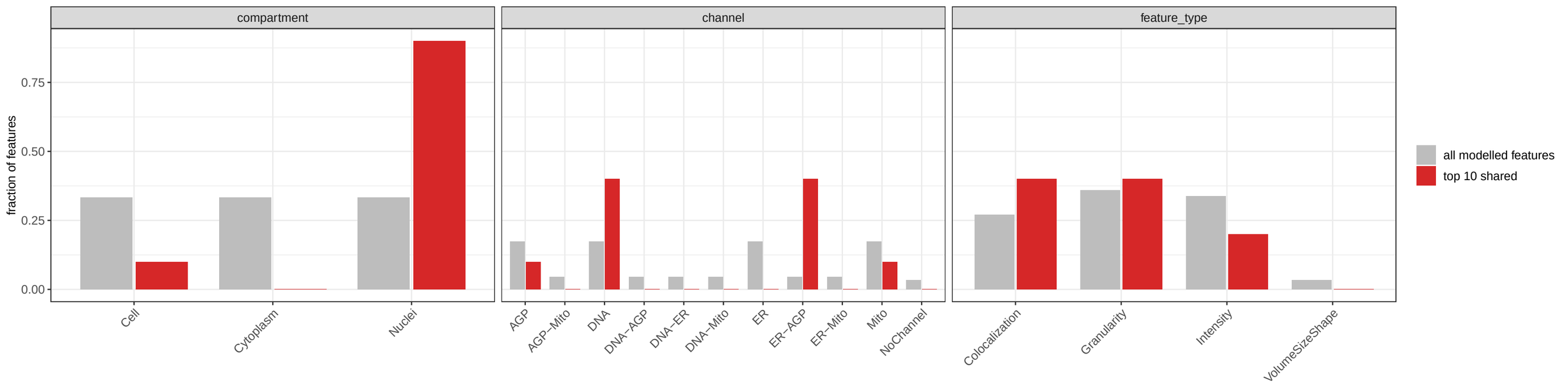

46
24
AGP
22
AGP−Mito
47
39
DNA
16
DNA−AGP
23
DNA−ER
20
DNA−Mito
48
35
ER
19
ER−AGP
22
ER−Mito
48
41
Mito
8
NoChannel
Granularity
Intensity
Colocalization
VolumeSizeShape
distinct features
50
40
30
20
10

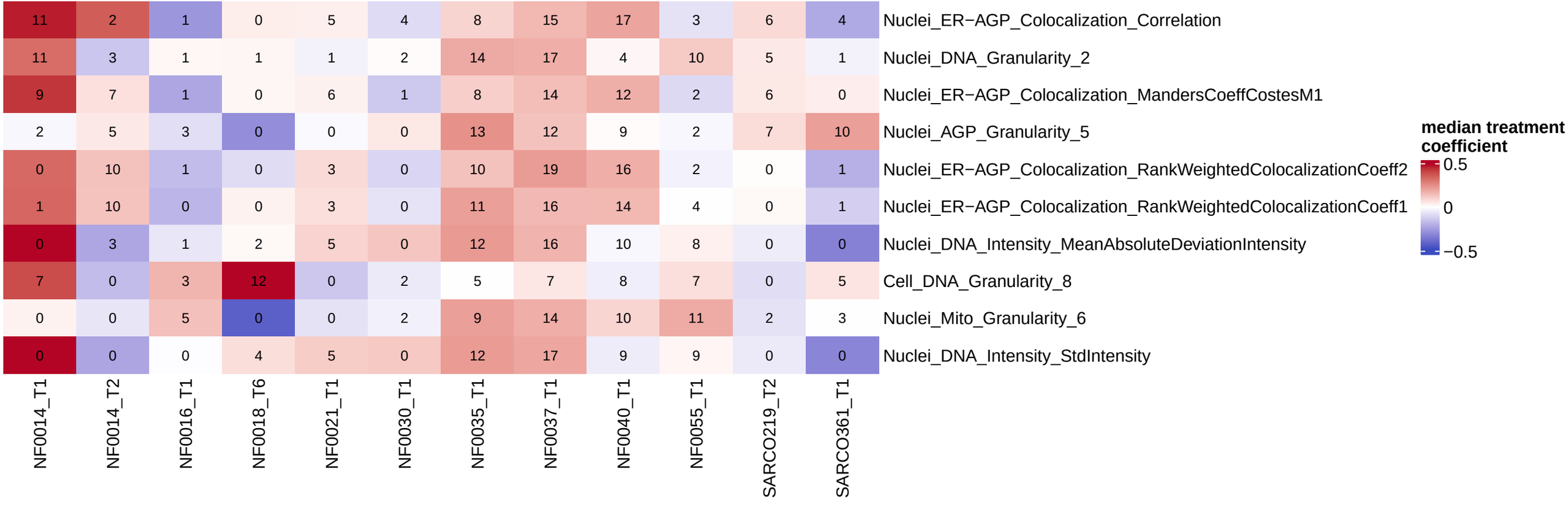
Nuclei_ER−AGP_Colocalization_Correlation
Nuclei_DNA_Granularity_2
Nuclei_ER−AGP_Colocalization_MandersCoeffCostesM1
Nuclei_AGP_Granularity_5
Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff2
Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff1
Nuclei_DNA_Intensity_MeanAbsoluteDeviationIntensity
Cell_DNA_Granularity_8
Nuclei_Mito_Granularity_6
Nuclei_DNA_Intensity_StdIntensity
Cell_DNA_Granularity_5
Cell_Mito_Granularity_11
Nuclei_DNA−ER_Colocalization_RankWeightedColocalizationCoeff2
Cell_Mito_Granularity_6
Nuclei_DNA_Granularity_3
Cell_NoChannel_VolumeSizeShape_Extent
Nuclei_DNA_Granularity_6
Cell_DNA_Granularity_7
Nuclei_DNA_Granularity_5
Nuclei_Mito_Granularity_8
Cytoplasm_DNA_Granularity_7
Nuclei_DNA_Granularity_4
Cell_Mito_Granularity_4
Nuclei_Mito_Granularity_14
Nuclei_Mito_Intensity_MassDisplacement
Cell_Mito_Granularity_7
Nuclei_DNA−Mito_Colocalization_MandersCoeffCostesM2
Nuclei_Mito_Granularity_4
Cytoplasm_DNA_Granularity_8
Cytoplasm_Mito_Granularity_7
Cytoplasm_Mito_Granularity_11
Nuclei_Mito_Granularity_7
Cytoplasm_AGP_Granularity_8
Nuclei_Mito_Granularity_10
Cytoplasm_AGP_Granularity_9
Nuclei_Mito_Granularity_2
Cytoplasm_DNA_Granularity_4
Cytoplasm_DNA_Granularity_9
Nuclei_Mito_Granularity_5
Nuclei_Mito_Granularity_11

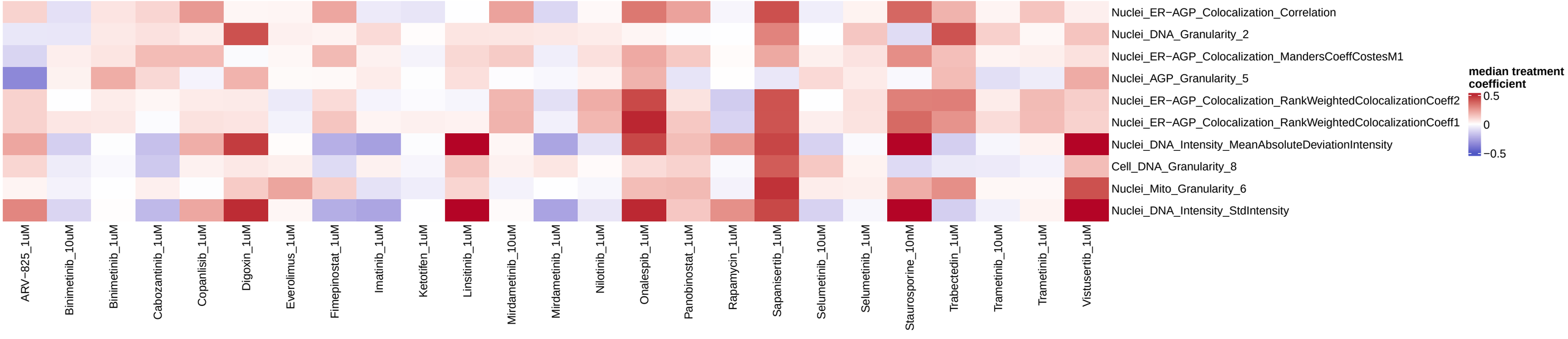
ARV−825_1uM
Binimetinib_10uM
Binimetinib_1uM
Cabozantinib_1uM
Copanlisib_1uM
Digoxin_1uM
Everolimus_1uM
Fimepinostat_1uM
Imatinib_1uM
Ketotifen_1uM
Linsitinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Nilotinib_1uM
Onalespib_1uM
Panobinostat_1uM
Rapamycin_1uM
Saparisertib_1uM
Selumetinib_10uM
Selumetinib_1uM
Staurosporine_10nM
Trabectedin_1uM
Trametinib_10uM
Trametinib_1uM
Vistusertib_1uM

patients (feature is treatment−only)
8
6
4
2
0


Nuclei_ER−AGP_Colocalization_Correlation
Nuclei_DNA_Granularity_2
Nuclei_ER−AGP_Colocalization_MandersCoeffCostesM1
Nuclei_AGP_Granularity_5
Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff2
Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff1
Nuclei_DNA_Intensity_MeanAbsoluteDeviationIntensity
Cell_DNA_Granularity_8
Nuclei_Mito_Granularity_6
Nuclei_DNA_Intensity_StdIntensity
−2
0
2
4
6
treatment coefficient (all patient x treatment models)
treatment−only significant
FALSE
TRUE
0.0
0.1
0.2
0.3
fraction of models that are treatment−only significant


compartment
channel
feature_type
fraction of features
0.00
0.25
0.50
0.75
Cell
Cytoplasm
Nuclei
AGP
AGP−Mito
DNA
DNA−AGP
DNA−ER
DNA−Mito
ER
ER−AGP
ER−Mito
Mito
NoChannel
Colocalization
Granularity
Intensity
VolumeSizeShape
all modelled features
top 10 shared


11 2 1 0 5 4 8 15 17 3 6 4 Nuclei_ER−AGP_Colocalization_Correlation
11 3 1 1 1 2 14 17 4 10 5 1 Nuclei_DNA_Granularity_2
9 7 1 0 6 1 8 14 12 2 6 0 Nuclei_ER−AGP_Colocalization_MandersCoeffCostesM1
2 5 3 0 0 0 13 12 9 2 7 10 Nuclei_AGP_Granularity_5
0 10 1 0 3 0 10 19 16 2 0 1 Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff2
1 10 0 0 3 0 11 16 14 4 0 1 Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff1
0 3 1 2 5 0 12 16 10 8 0 0 Nuclei_DNA_Intensity_MeanAbsoluteDeviationIntensity
7 0 3 12 0 2 5 7 8 7 0 5 Cell_DNA_Granularity_8
0 0 5 0 0 2 9 14 10 11 2 3 Nuclei_Mito_Granularity_6
0 0 0 4 5 0 12 17 9 9 0 0 Nuclei_DNA_Intensity_StdIntensity
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
median treatment coefficient
0.5
0
−0.5


Nuclei_ER−AGP_Colocalization_Correlation
Nuclei_DNA_Granularity_2
Nuclei_ER−AGP_Colocalization_MandersCoeffCostesM1
Nuclei_AGP_Granularity_5
Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff2
Nuclei_ER−AGP_Colocalization_RankWeightedColocalizationCoeff1
Nuclei_DNA_Intensity_MeanAbsoluteDeviationIntensity
Cell_DNA_Granularity_8
Nuclei_Mito_Granularity_6
Nuclei_DNA_Intensity_StdIntensity
median treatment coefficient
0.5
0
−0.5
ARV−825_1uM
Binimetinib_10uM
Binimetinib_1uM
Cabozantinib_1uM
Copanlisib_1uM
Digoxin_1uM
Everolimus_1uM
Fimepinostat_1uM
Imatinib_1uM
Ketotifen_1uM
Linsitinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Nilotinib_1uM
Onalespib_1uM
Panobinostat_1uM
Rapamycin_1uM
Sapanisertib_1uM
Selumetinib_10uM
Selumetinib_1uM
Staurosporine_10nM
Trabectedin_1uM
Trametinib_10uM
Trametinib_1uM
Vistusertib_1uM